

ARIANTO BAGUS

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Sidoarjo, East Java

Physics graduate (B.Sc.) from Institut Teknologi Sepuluh Nopember (ITS) specializing in acoustics and instrumentation, with 3+ years of experience as a laboratory assistant in Experimental Physics and Acoustic Instrumentation. Hands-on skills in operating laboratory equipment, collecting and analyzing experimental data using MATLAB and Python, and applying laboratory safety procedures. Experience in lecturer-led research on a railway weigh-in-motion system and in building an Arduino-based hydroponic monitoring and control system. Currently a B2B Sales Engineer providing technical consultation and product support to industrial clients. A fast learner who works well independently and in a team.

Education

Institut Teknologi Sepuluh Nopember - Surabaya — Jul 2019 - Feb 2024
Bachelor of Science in Physics, 3.14/4.00

- Research focus: acoustic physics and theoretical physics
- Final project: "Analysis of the Effect of Finger Position on Acoustic Guitar Fretting Technique on Sound Quality based on Mersenne's Law", using a microcontroller-based system to determine finger positions that produce target frequencies

Work Experience

Perum BULOG - Surabaya (Remote) — Jun 2026 - Sep 2026
Data Verification Officer (Food Assistance Program) - Freelance
Indonesian state-owned logistics enterprise responsible for managing national food supply and distributing government food assistance

- Verified distribution data of government food assistance to ensure accuracy and completeness
- Cross-checked beneficiary records against reports and flagged discrepancies
- Compiled verification results and reported findings to the supervising team

PT. Tata Rapika Globalindo - Sidoarjo — Aug 2025 - Present
B2B Sales Engineer - Full Time
Sheet metal fabrication and engineering company in Sidoarjo serving industrial and business clients

- Provide technical consultation and product recommendations to business clients based on their application requirements
- Prepare technical proposals, quotations, and product presentations
- Coordinate with internal technical teams to support project delivery and after-sales needs

Perum BULOG - Surabaya — Jun 2025 - Sep 2025
Food Assistance Distribution Officer - Contract

- Distributed government food assistance to beneficiaries according to schedules and procedures
- Recorded distributions accurately and coordinated with warehouse and field teams

SMA Negeri 3 Sidoarjo - Sidoarjo — Jul 2019 - Jun 2025
Robotics Instructor - Freelance
School robotics extracurricular established in 2013, with electrical, mechanical, and programming divisions. Actively competes from regional to national levels

- Taught robotics fundamentals across electrical, mechanical, and programming topics
- Coached students for regional and national competitions
- Student achievements: Best Ship Design at Automation Week 2019; 1st Place at ARA ITS 4.0 Cyber Security Olympiad 2022

Uno Education - Surabaya — Oct 2023 - Jul 2024
Tutor - Contract
Tutoring institution established in 2008, preparing students and exam candidates for national and international competitions

- Mentored 5 high school students for math and physics olympiads, with 3 winning awards

Laboratory & Research Experience

Institut Teknologi Sepuluh Nopember (ITS) - **Surabaya** Jan 2023 - Oct 2023

Acoustic Instrumentation Physics Laboratory Assistant

ITS Physics Department selects active students into one of its six focused laboratory fields, including Acoustic Instrumentation

- Assisted students in conducting research using laboratory equipment
- Operated laboratory instruments to collect and process experimental data
- Supported lecturer-led acoustic research projects

Institut Teknologi Sepuluh Nopember (ITS) - **Surabaya** May 2020 - Jun 2023

Experimental Physics Laboratory Assistant

ITS selects students as laboratory assistants for its physics laboratories, including Basic Physics I & II, Classical Physics, and Modern Physics

- Assisted practical sessions for Fundamental Physics I & II, Classical Physics, and Modern Physics
- Presented practical material to students and enforced laboratory safety (K3) procedures

Institut Teknologi Sepuluh Nopember - **Surabaya** Jan 2021 - Jul 2022

Research Team Member & Technician (Lecturer-led project)

Lecturer-led project to develop a system that measures the weight of moving trains using load cells mounted on the rail and the shift in the rail's natural frequency

- Designed the mechanical structure for mounting load cells on the rail and integrated it with the electrical system
- Built the electronic circuits and sensor interfaces, and programmed the microcontroller for data acquisition
- Applied vibration-based analysis, in which the train's mass lowers the rail's natural frequency, to determine total train weight
- Calibrated and tested the system and analyzed the measurement data

KebonID - **Appropriate Technology Hydroponic Startup - Surabaya** Mar 2017 - Jul 2020

Electrical Technician (Founder)

Small startup that grew out of an appropriate technology project, which won at LKTTG 2017 Sidoarjo (Regional Appropriate Technology Competition). Operations ended in 2020 due to limited funding

- Designed the circuitry and wiring for an Arduino-based hydroponic system that monitors nutrient levels, temperature, and humidity
- Installed and calibrated sensors, and assembled actuators (nutrient/water pump and pH up/down pumps) that keep growing conditions optimal automatically
- Programmed the Arduino for sensor reading and automatic actuator control

Organisational Experience

ITS Foreign Language Society (UKM IFLS ITS) - **Surabaya** Jan 2020 - Dec 2022

Head of Division

Student activity unit at ITS focused on foreign languages and culture, comprising the ITS Debating Society, Japan, Korea, Dueling Cards, and IFLS Culture divisions

- Led one of five interest divisions with 3 sub-divisions and 500+ members
- Managed program schedules and routine division activities, and conducted monthly evaluations of each department
- Supervised two major annual events (INOCHI and K-Fest) with 1,000+ visitors per day

Himpunan Mahasiswa Fisika ITS (Himasika ITS) - **Surabaya** Jul 2021 - Jan 2022

Staff, Instructure Committee

Student organization of the ITS Physics Department

- Organized the "Study Orientation" event for new Physics Department students, handling on-stage program and backstage coordination
- Designed the event concept around moral and educational values

Skills

Laboratory & Technical: Laboratory equipment operation · Instrumentation · Experimental data collection and analysis · Experimental physics · Acoustics and acoustic data analysis · Laboratory safety (K3) · Sensor integration and calibration · Circuit and mechanical design for instrumentation

Software & Programming: MATLAB · Python · Arduino / Microcontrollers · Microsoft Office

Languages: Indonesian (Native), English (TEFL ITS score: 487, 2024)

Soft Skills: Technical consultation · Fast learning and quick adaptation · Teamwork and independent work · Able to perform under pressure